CSE 2025 ll Rancho Bernardo High School

Water Quality

Name(s)_____

Score: __/40

Section 1: Freshwater ecology

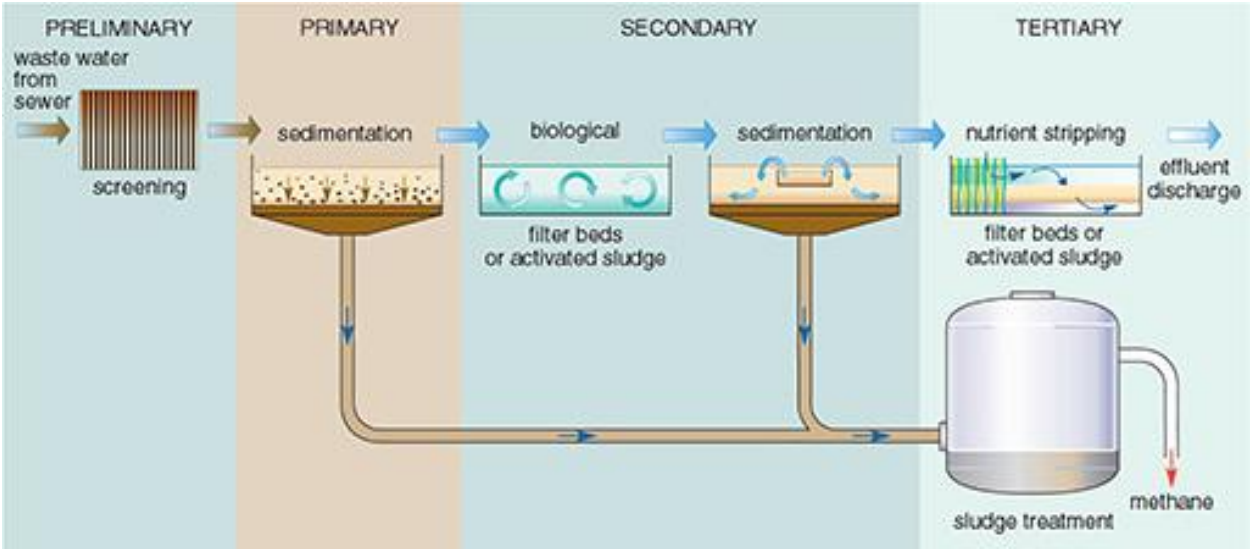
1. a.
2. Temperature, Salinity, Pressure, Biological Activity
3. c.
4. Gasoline, oil, industrial wastes, urban runoff
5. d.
6. b.
7. a.
8. Dysentery, Typhoid Fever, Gastroenteritis, Hepatitis
9. a.
10. biochemical oxygen demand (BOD). Measures how fast bacteria use up the oxygen in the water to decompose organic matter/ decompose organic pollution.
11. Increase in water temperature, decrease of photosynthetic rate, decreased growth, more aesthetically displeasing water, reduce the ability of fish gills to absorb dissolved oxygen, increased risk of development of gastrointestinal diseases, shield bacteria from certain types of sterilization of water

Section 2: Macroflora and Fauna Identification

1. Stonefly
2. Plecoptera
3. fly over water and drop the eggs into the water or hang them to rocks or branches
4. diapause
5. moulting
6. cranefly
7. common in northern latitudes and high elevations, warm temperature, high humidity, abundant rainfall
8. Moderately clean water
9. Few days to 2 weeks
10. water boatman
11. zoophagy
12. Trap air underneath their wings and breathe from it
13. make chirping sound by rubbing their long forelegs on their head
14. Water hyacinth
15. Primarily by runners or stolons which lead to daughter plants
16. Azotobacter chroococcum
17. Forms dense layer on surface, blocks sunlight, impedes water flow, outcompete native plant species

Section 3: Analysis

1-3.



4. Can cause overpopulation of plants, increase plant growth rates, lower DO, and decrease primary productivity.
5. Manatee, Egeria Densa, Parrot's Feather (Myriophyllum aquaticum), brook trout, and there can be more answers to this question.
6. pH is 5.8 and no the water cannot support life
7. B
8. C